

Table S: **Interaction quality in CookSim.** Unaggregated scores for layout review.

Assistant	Overall	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness
Frontier API models						
GPT-6 Astra	64	87	89	79	97	86
Gemini 3.1 Pro	28	61	66	57	80	65
Gemini 3.8 Flash	53	83	80	68	76	89
Real-time streaming API models						
Gemini 3.8 Live (polled)	47	81	75	61	72	91
GPT-Realtime 2.1 (polled)	41	82	78	66	71	53
Gemini 3.8 Live (native)	40	75	78	58	78	86
GPT-Realtime 2.1 (native)	50	81	88	73	76	58
Open-weight non-streaming VLMs						
Qwen3.8-27B	37	71	82	62	86	72
Molmo2-8B	12	54	32	32	33	19
InternVideo3-8B	21	66	45	35	42	69
GLM-4.6V-Flash	47	90	59	54	62	82
Kimi-VL-A3B-Thinking	6	34	39	29	39	26
Muse Glimmer	27	59	75	57	65	41
Open-weight streaming VLMs						
MiniCPM-o 4.5	15	55	45	33	46	52
Qwen3-Omni-30B	25	66	64	51	54	68
Proact-VL	16	66	26	17	28	45
StreamingVLM	24	76	25	35	34	23

Scores are on a 0–100 scale (run 1). Overall: full 16-item CookSim rubric with mandatory-failure gating, not the mean of the five displayed dimensions. Dimension scores are averaged within episodes and then across episodes. Bold: highest score per column.